HOSPITAL PERFORMANCE ANALYTICS DASHBOARD

DASHBOARD TESTING REPORT

Project Report

Submitted by

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Technology Used

Tableau

Dashboards Tested

Hospital Overview
Patient Flow
Department Analysis
Resource Utilization

Testing Activities

KPI Calculation Validation
Healthcare Metrics Verification
Dashboard Interaction Testing
Patient-Flow Analytics Validation

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1 Introduction

The Hospital Performance Analytics Dashboard is developed to present important hospital performance indicators in a clear and interactive format. The dashboard enables users to examine admissions, patient flow, department-level information, and resource utilization through visual analytics.

Testing is an essential stage of dashboard development because it helps identify calculation errors, inconsistent metric definitions, unexpected interactions, and incorrect analytical results.

This report documents the testing performed on the Hospital Performance Analytics Dashboard using Tableau. The testing covers KPI calculations, healthcare metric definitions, dashboard interactions, and patient-flow analytics.

1.1 Objectives

The main objectives of the testing process are:

1. To validate the calculations used in the KPI cards.
2. To verify the definitions and interpretation of healthcare performance metrics.
3. To test dashboard navigation, filters, and chart actions.
4. To validate patient counts, average waiting time, and average patient age.
5. To check consistency between related dashboard views.
6. To confirm that the dashboard produces the expected results for the tested conditions.

1.2 Scope

The testing covers the following four dashboards:

- Hospital Overview
- Patient Flow
- Department Analysis
- Resource Utilization

The scope includes the visible KPI values, calculation logic, healthcare metric interpretation, dashboard controls, and patient-flow visualizations.

2 Testing Methodology

The dashboard was tested by examining the displayed KPIs, checking the associated calculations against the project's data and metric definitions, and interacting with the dashboard controls.

The testing process included the following activities:

1. Opening each dashboard and reviewing the displayed metrics and visualizations.
2. Checking the calculation logic and aggregation used for the relevant KPI fields.
3. Comparing calculated results with the displayed dashboard values under matching filter conditions.
4. Changing available filters and checking the resulting changes in charts, tables, and KPI cards.
5. Testing navigation buttons and configured chart actions.
6. Reviewing patient-flow metrics and their underlying definitions.
7. Recording the observed results for each test case.

2.1 Testing Criteria

A test case was considered successful when the observed result matched the expected result for the data, filters, and calculation definition used in the test.

Calculations were evaluated using the relevant source data and the project's intended metric definitions. Dashboard interactions were evaluated by checking whether the configured controls produced their expected behaviour.

3 KPI Calculation Validation

KPI calculation validation was performed to check whether the summary values displayed on the Hospital Overview and Patient Flow dashboards were consistent with their calculation logic and the underlying data.

3.1 Recorded KPI Values

The following values were recorded from the dashboard views during the project testing process.

Dashboard	**KPI**	**Value**	**Validation Performed**
Hospital Overview	Total admissions	7,985	Compared with the admission count under the same filter conditions.
Hospital Overview	Occupancy rate	81.83	Checked the calculation and interpretation of the occupancy measure.
Hospital Overview	Average length of stay	15.59	Checked the stay-duration calculation and its unit.
Hospital Overview	Readmission rate	0.4147	Checked the readmission definition, denominator, and displayed format.
Hospital Overview	Bed utilization	81.83	Reviewed the calculation and compared it with the occupancy measure.
Patient Flow	Patient count	7,985	Checked the count using the same applicable filter conditions.
Patient Flow	Average waiting time	67.88	Checked the waiting-time calculation and its unit.

Dashboard	KPI	Value	Validation Performed
Patient Flow	Average patient age	47.19	Checked the age aggregation and handling of eligible records.
Department Analysis	Neurology referrals	2,796	Checked the department grouping and referral aggregation.
Department Analysis	Gastroenterology referrals	1,811	Checked the department grouping and referral aggregation.
Resource Utilization	Yearly trend	2020–2024	Checked the displayed year range and resource-related aggregation.

3.2 Calculation Checks

3.2.1 Total Admissions

The total admissions KPI was checked against the admission records using the same filter conditions. The counting method was reviewed to ensure that the result corresponded to the intended admission measure.

3.2.2 Occupancy Rate

The occupancy calculation was reviewed using the metric definition applied in the project. Where occupancy is defined as occupied beds divided by available beds, the percentage is calculated as:

$$OccupancyRate = \frac{OccupiedBeds}{AvailableBeds} \times 100$$

The reporting period and bed-count definitions were considered when comparing the calculated result with the dashboard value.

3.2.3 Average Length of Stay

Average length of stay was checked using the stay-duration measure defined for the project. Admission and discharge information, eligible records, and the unit of measurement were considered during validation.

$$AverageLengthofStay = \frac{TotalStayDuration}{NumberofEligibleAdmissions}$$

3.2.4 Readmission Rate

The readmission calculation was reviewed using the project's definition of a readmission and its eligible population. The numerator, denominator, time window, and display format were checked to ensure that the reported value was interpreted correctly.

$$ReadmissionRate = \frac{EligibleReadmissions}{DefinedEligiblePopulation} \times 100$$

3.2.5 Average Waiting Time

The average waiting-time KPI was checked against the waiting duration defined by the project. The timestamp fields used to calculate waiting time and the unit of measurement were reviewed.

$$AverageWaitingTime = \frac{TotalValidWaitingDuration}{NumberofEligibleVisits}$$

3.2.6 Average Patient Age

The average patient age was reviewed using the age field and the applicable patient records. The aggregation and treatment of missing or invalid age values were considered.

4 Healthcare Metrics Verification

Healthcare metric verification was carried out to ensure that the displayed indicators were meaningful, consistently defined, and suitable for interpreting hospital performance.

4.1 Metric Definition Checks

The following aspects were reviewed during metric verification:

- The meaning and purpose of each healthcare metric.
- The numerator and denominator used in rate calculations.
- The unit used for waiting time and length of stay.
- The time period and filters applied to the metric.
- The aggregation method used for counts and averages.
- The consistency of metric definitions across dashboards.
- The handling of missing values and invalid records.

4.2 Occupancy Rate and Bed Utilization

Occupancy rate and bed utilization were reviewed as separate measures because their definitions can differ. Occupancy may represent the proportion of available beds occupied at a particular time, whereas bed utilization may be calculated using occupied bed-days over available bed-days.

The two values displayed on the dashboard were compared with their respective definitions to determine whether the matching values were expected.

4.3 Readmission Rate Interpretation

The readmission rate value displayed on the dashboard was reviewed to ensure that the calculation and number formatting were consistent. A decimal value should not automatically be interpreted as a percentage without confirming the calculation and formatting applied in Tableau.

4.4 Metric Consistency

Related metrics were compared across dashboard views using matching filter conditions. The purpose was to identify differences caused by inconsistent counting rules, aggregation methods, or filter configurations.

5 Dashboard Interaction Testing

Dashboard interaction testing was conducted to evaluate navigation, filtering, chart actions, and reset behaviour.

5.1 Navigation Testing

The navigation controls were tested to verify that each dashboard opened the intended view.

- Hospital Overview
- Patient Flow
- Department Analysis
- Resource Utilization

The navigation behaviour was checked for correct destination and consistent dashboard presentation.

5.2 Filter Testing

Available filters were changed individually to observe their effects on KPI cards, charts, and tables. The resulting values were checked to ensure that the relevant views responded according to the configured filter behaviour.

5.3 Chart Action Testing

Interactive chart marks were selected to check configured highlight, filtering, and tooltip behaviour. The displayed results were reviewed to ensure that interactions corresponded to the selected data points.

5.4 Reset Behaviour

Where reset or clear-filter controls were available, they were tested to check whether the dashboard returned to its intended default state.

5.5 Interaction Test Cases

Test ID	Test Case	Expected Result	Result
INT-01	Navigation	Each navigation control opens its intended dashboard.	Pass
INT-02	Filter response	Relevant KPIs and visualizations update according to the selected filters.	Pass
INT-03	Chart actions	Configured chart selections and actions behave as expected.	Pass
INT-04	Reset behaviour	Available reset controls restore the intended default filter state.	Pass

6 Patient-Flow Analytics Validation

Patient-flow analytics were reviewed to check the consistency of patient counts, average waiting time, and average patient age. The related visualizations were also examined under different applicable filter conditions.

6.1 Patient Count Validation

The patient count displayed on the Patient Flow dashboard was compared with the corresponding source records. Matching filter conditions and the project's counting method were used during the comparison.

The count was also compared with the Hospital Overview admissions value after considering whether both measures represented the same population and counting rule.

6.2 Waiting-Time Validation

The waiting-time measure was checked against the timestamps and duration definition used in the project. The validation considered the following aspects:

- The start and end timestamps used in the calculation.
- The unit in which waiting time is represented.
- Missing or invalid timestamp values.
- Negative or implausible waiting durations.
- The effect of available filters on the displayed average.

6.3 Average Patient Age Validation

The average patient age was reviewed using the relevant age field and eligible records. The aggregation method, missing values, and consistency of the displayed average were checked.

6.4 Patient-Flow Interaction Validation

The available filters were applied to the Patient Flow dashboard to check whether the related charts, patient counts, and summary values responded as expected. The resulting visualizations were reviewed for consistency with the selected filter conditions.

6.5 Patient-Flow Test Cases

Test ID	Test Case	Expected Result	Result
FLOW-01	Patient count	The displayed count matches the applicable source records and counting rule.	Pass
FLOW-02	Average waiting time	The waiting-time average follows the defined duration calculation and unit.	Pass
FLOW-03	Average patient age	The displayed average follows the age aggregation and eligibility rules.	Pass
FLOW-04	Filter response	Patient-flow visualizations respond correctly to applicable filters.	Pass

7 Overall Test Results

The testing process covered KPI calculations, healthcare metric definitions, dashboard interactions, and patient-flow analytics. The results recorded for the completed test cases are summarized below.

7.1 Test Summary

S. No.	Testing Area	Recorded Result
1	KPI calculation validation	Pass
2	Healthcare metrics verification	Pass
3	Dashboard interaction testing	Pass
4	Patient-flow analytics validation	Pass

7.2 Observations

The testing activities covered the displayed hospital KPIs, healthcare metric definitions, dashboard navigation, filter responses, chart actions, and patient-flow indicators.

The dashboard values and interactions were reviewed using the test conditions applied during validation. Metric units, calculation definitions, and filter consistency were considered when interpreting the results.

The recorded test outcomes indicate that the tested dashboard functions met their expected results under the conditions used during testing.

8 Conclusion

The Hospital Performance Analytics Dashboard was tested using Tableau across four major areas: KPI calculation validation, healthcare metric verification, dashboard interaction testing, and patient-flow analytics validation.

The testing process reviewed the main performance indicators, including total admissions, occupancy rate, average length of stay, readmission rate, bed utilization, average waiting time, and average patient age. It also covered dashboard navigation, filter behaviour, chart interactions, and consistency between related visualizations.

Based on the results recorded for the completed test cases, the dashboard met the expected outcomes for the checks performed. The testing process helped evaluate the accuracy, consistency, and usability of the dashboard for hospital performance analysis.

The completed report documents the testing approach, observed KPI values, validation procedures, test cases, and overall results for the project.

Final Status

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Final Status	Passed

End of Report